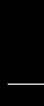


LightningPiggy & The Unveiling of MicroPythonOS



LightningPiggy



Teach kids when they're young



- Fun
- Personal
- Save
- Invest
- Bitcoin and Sats

Why not a smartphone?



- Fragile
- Complex
- Expensive
- Distractions
- Turns Off

Perfect Kid's Wallet



- Always On (ePaper)
- Balance in Sats
- Amount & Comment
- Receive QR code
- Instant refresh

Proof-of-Concept



- Credit: BlackCoffee
- LilyGo ePaper Board
- Backend: LNBits
- Hard-coded Config
- Arduino IDE Installer

Awesome Added Features



- Easy Web Installer
- Battery Management
- Shake-to-Wake
- Update Notifications
- Webbrowser Config

Great Product Finished



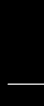
- Wisdom and Greeting
- Fiat Balance & Price
- Localization
- Access Point Config
- Nostr Wallet Connect

Build Your Own

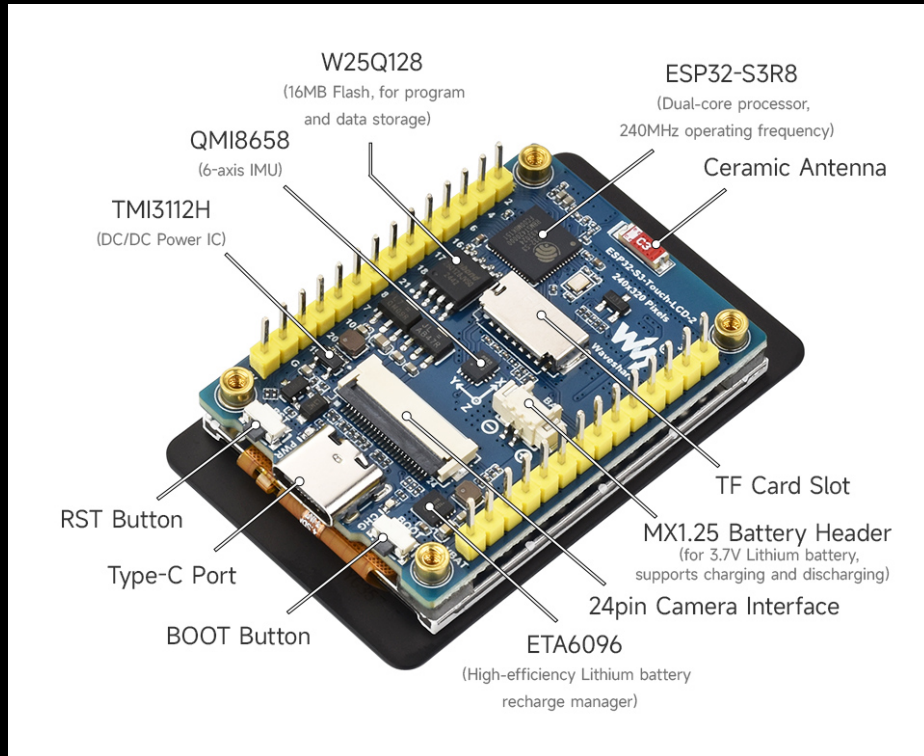


- LightningPiggy.com
 - /market
 - /build
- Make It Your Own
- 3D Printable Cases
- Valuable Present

New LightningPiggy



New LightningPiggy



- Color Touch Screen
- 251% More Pixels
- Camera
- Accelerometer
- MicroSD Card

Need Android-like OS for ESP32



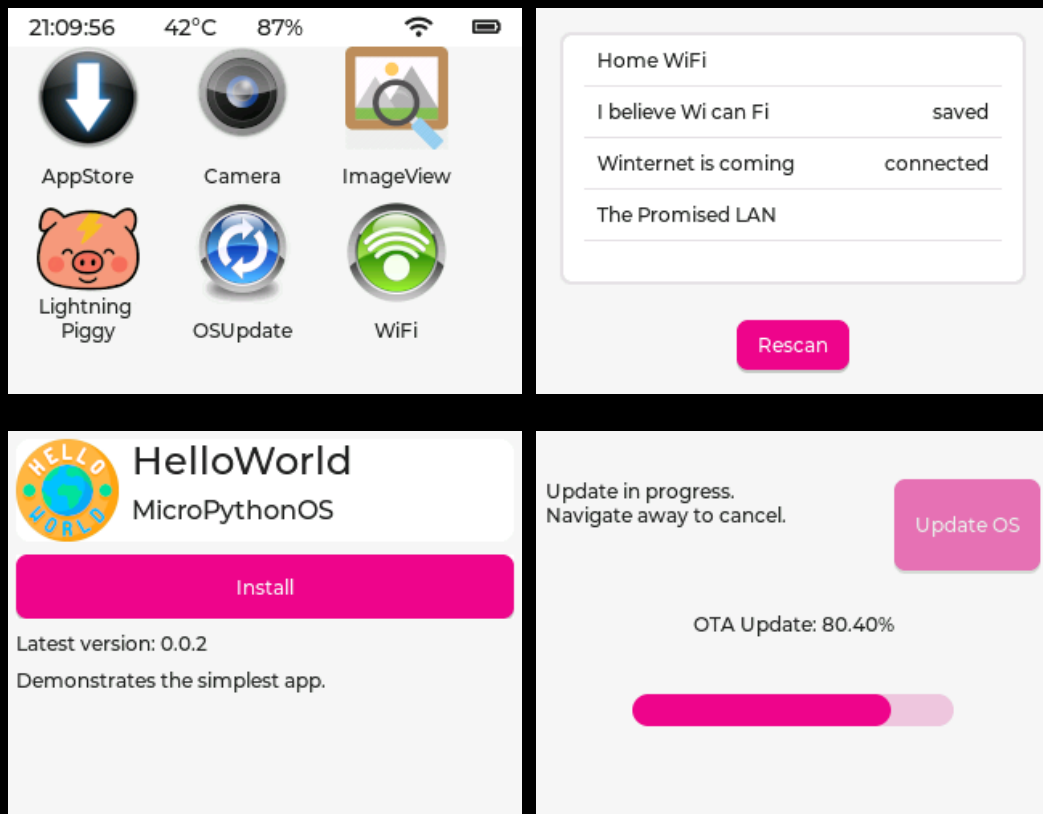
- WiFi Manager
- Update and Rollback
- Slick User Interface
- Easy Development
- App Store

Impossible?



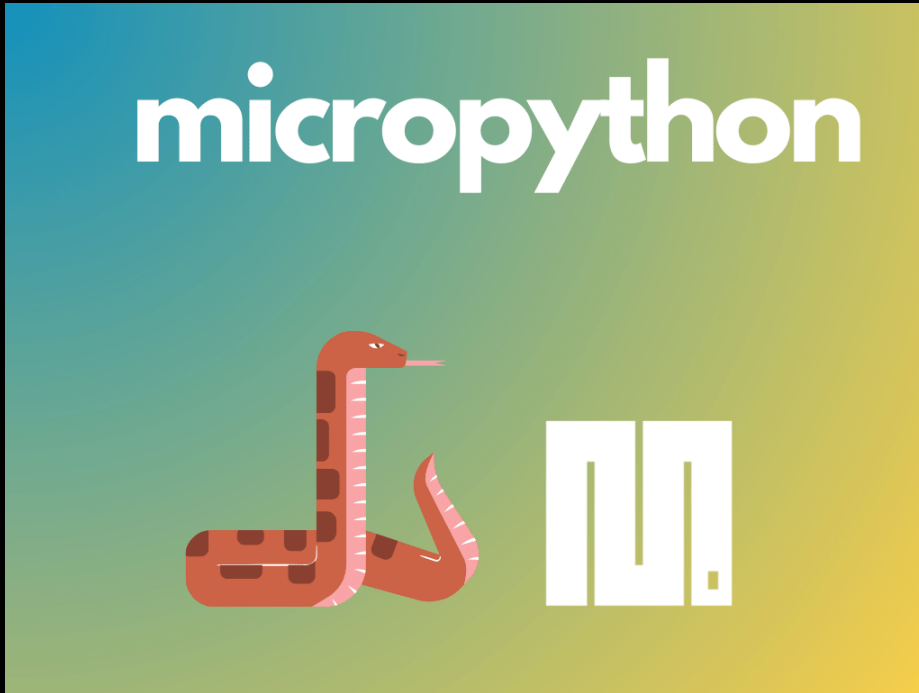
- Like IBM PS/2 (1981)
- No MMU
- 300KB SDRAM
- 8MB PSRAM
- 16MB ROM

It's MicroPythonOS!



- Launcher
- WiFi + Auto-Connect
- OSUpdate
- AppStore
- Settings

MicroPython Works



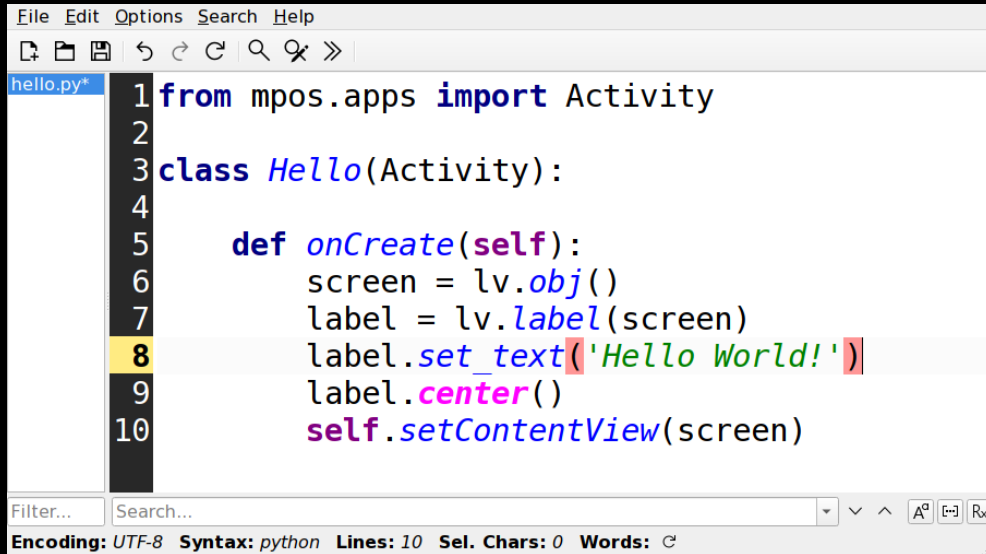
- Born 2014, ESP 2017
- Surprisingly fast
- Also desktop (native)
- Quick Coding Cycle
- `mip.install()`

LVGL 9.3



- Lightweight
- 30+ Widgets
- Animations
- Image Decoders
- L10n, i18n
- UI Editor

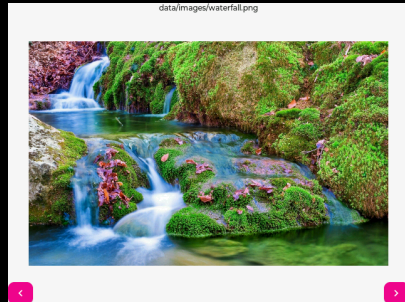
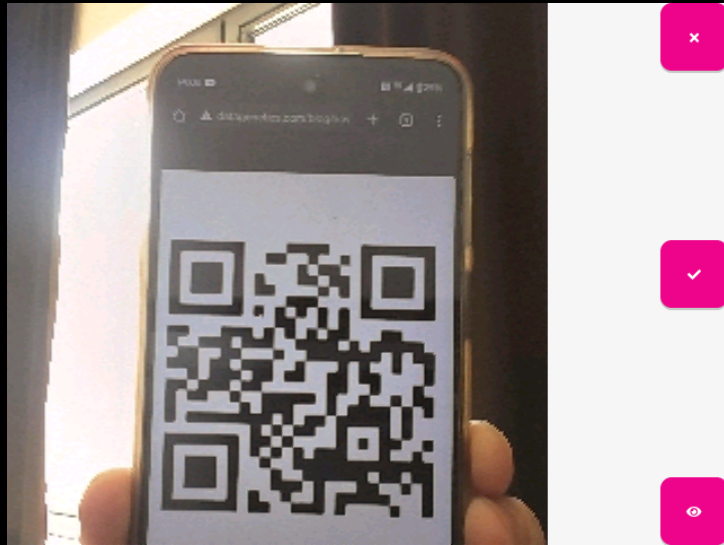
Easy Android-like Development Flow

A screenshot of a code editor window titled 'hello.py*'. The editor has a menu bar with 'File', 'Edit', 'Options', 'Search', and 'Help'. Below the menu is a toolbar with icons for file operations and navigation. The code is written in Python and defines a class 'Hello' that inherits from 'Activity'. The 'onCreate' method is implemented, where it creates a screen, adds a label, sets the label text to 'Hello World!', centers the label, and sets the content view to the screen. The line numbers 1 through 10 are visible on the left side of the code. The status bar at the bottom shows 'Encoding: UTF-8', 'Syntax: python', 'Lines: 10', 'Sel. Chars: 0', and 'Words: 0'.

```
1 from mpos.apps import Activity
2
3 class Hello(Activity):
4
5     def onCreate(self):
6         screen = lv.obj()
7         label = lv.label(screen)
8         label.set_text('Hello World!')
9         label.center()
10        self.setContentView(screen)
```

- Create Screen
- Add Label
- Set Label Text
- Center Label
- Activate Screen

More Apps in the AppStore



- Camera + QR decode
- Image Viewer
- <Your Apps Here>
 - Games / Education
 - Nostr Client
 - Music Player
- Lightning Piggy

LightningPiggy App

164 sats

8 sats: 8 coins
2 sats: 2 coins
7 sats: seven
6 sats: up
5 sats: balance up
4 sats: sweet!



Wallet Type

- ☐ LNBits
☒ Nostr Wallet Connect

Save

Cancel

Nostr Wallet Connect

nostr+walletconnect://ba80990666e...

Save

Cancel

Scan data from QR code

Tip: Create your own QR code,
using <https://genqrcode.com> or another tool.

176 sats

12 sats: 12
8 sats: 8 coins
2 sats: 2 coins
7 sats: seven
6 sats: up
5 sats: balance up



- First Release
- Showcase OS
- Validate Frameworks
- Feature Parity
- Ported python-nostr

Complementary LightningPiggies

- Grayscale ePaper
- Button and Tilt Input
- Receive Only
- Low Power Draw
- Single Purpose

- Color IPS Panel
- Touch Screen Input
- Scan QR to Spend
- High Power Draw
- Multi Purpose

Live Demo



- You are looking at it.
- Yes, really!
- You were in it...
- the whole time!

MicroPythonOS.com is Here



- Open Source
- Open to Developers
- Perfect for OEM's
- Grants and Funding
- Commercialization

Your turn!



**WE WANT YOU
IN OUR
TEAM!**

- Get a few Piggies
- Build Cool Apps
- Share
- Reach out
- t.me/MicroPythonOS